

Benjamin Bubnick

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EDUCATION

UNIVERSITY OF CINCINNATI MS IN PHYSICS

2008 - 2010 | Cincinnati, OH
Thesis: Massive Stellar Clusters
in the Disk of the Milky Way Galaxy
Cum. GPA: 3.313 / 4.0

OHIO UNIVERSITY BACHELOR OF FINE ARTS

2000 - 2004 | Athens, OH
Thesis: A Novel Method for
Low-Toxic Photo-Etch Printing
Cum. GPA: 2.737 / 4.0
Major GPA: 3.554 / 4.0

LINKS

Personal:// BenBubnick.com
LinkedIn:// BenBubnick
Facebook:// Ben.Bubnick
Thesis:// Massive Clusters

COURSEWORK

MATHEMATICS

Statistics
Linear Algebra
Computational Modelling
Nonlinear Root-finding
Numerical Interpolation
Numerical Methods for
Integration & Differentiation

PHYSICS

Radiative Transfer
Fourier Analysis
Spectrophotometry
Infrared Spectroscopy
Astrophysics
(Teaching Assistant)
Introductory Physics

SKILLS

PROGRAMMING

.NET Framework:
VB • ASP • SQL • LINQ • C#
General:
HTML5 • IRAF • Mathematica
L^AT_EX • C++ • CSS • FORTRAN
IDL • JavaScript • JMP • LabVIEW
MATLAB • Python • Regex • Shell

WORK HISTORY

AMTRUST FINANCIAL SERVICES | SOFTWARE ENGINEER

2014 - Present | Cleveland, OH
Developed web forms for insurance industry using ISO standards and created multi-tier MVC components, Web APIs, and UIs using VB, ASP, Javascript, & HTML5

- Led development team on business integration of new acquisitions
- Implemented new unit testing method written in C#

PPG INDUSTRIES | COLOR SCIENTIST

2012 - 2013 | Strongsville, OH
Developed software and radiative transfer models on complex paint mixtures to determine formulation methods and for spectroscopic profile analysis on color match samples.

- 40% marked improvement on color matching software algorithm using numerical integration methods
- Developed cross-instrument correlation software for statistical risk analysis projects using JMP
- Debugged optimization function for numerical solution to a version of the radiative transfer equation written in VB.NET and FORTRAN
- Wrote DLL wrapper for on-loan spectroscopic instrument using Visual C++ and VB.NET

MIAMI UNIVERSITY | ADJUNCT PROFESSOR

2010 - 2011 | Hamilton, OH
Developed curricula in introductory physics & astronomy courses; shared time with Cincinnati State Community College

- Taught 3 courses: Astronomy, Introduction to Physics, and Physics & Society

RESEARCH

UC ASTROPHYSICS LAB | GRADUATE RESEARCH ASSISTANT

2008 - 2010 | Cincinnati, OH
Worked with **Prof Margaret Hanson** to study Main Sequence stellar infrared spectroscopy. Performed quality control for the Cambridge Astronomical Survey Unit Variable Stars database (Publication) and controlled the SpeX spectroscopic tool as a visiting astronomer to the NASA Infrared Telescope Facility (Publication)

AWARDS

2010	Mary J Hanna Fellowship	University of Cincinnati
2008 - 10	UGS Scholarship	University of Cincinnati
2003	Provost's Undergraduate Research Grant	Ohio University
2002 - 04,	Dean's List	Ohio University
2000,	Ohio Proficiency Scholarship	State of Ohio

SOCIETIES

2012 - 13	Detroit Color Council	
2009 - 11	American Institute of Aeronautics and Astronautics	Vice President
2008 - 10	University of Cincinnati Astronomy Club	Secretary
2005 - 08	Society of Physics Students	Secretary
2006 - 08	Sigma Theta Epsilon	Secretary